

Macroeconomics

Economic Growth

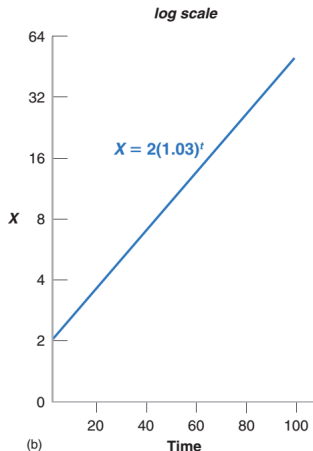
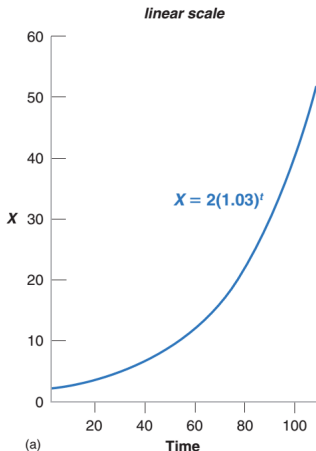
(Blanchard, Macroeconomics 8th edition Ch. 10-11)

Economic Growth

- ▶ Growth is the steady increase in aggregate output over time.
- ▶ We now shift our focus from economic fluctuations and the determination of output in the short and medium run ...
- ▶ to growth and the determination of output in the long run.
- ▶ Fluctuations fade when looking at activity over longer periods, e.g., decades.
- ▶ Some simple stylized facts will be presented.

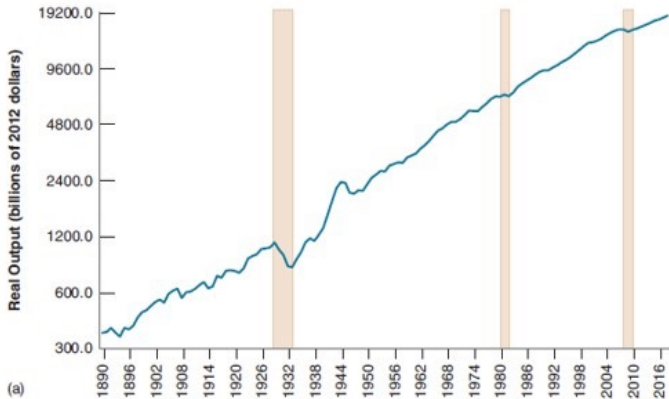
Logarithmic Scale

- Logarithm scale: the same proportional increase in a variable is represented by the same distance on vertical axis.



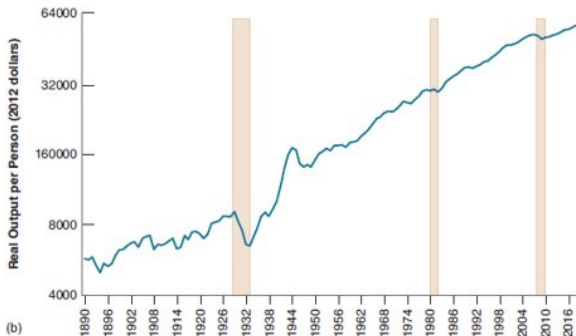
US GDP since 1890

- Enormous increase in US output since 1890, by a factor of 51.



US GDP per person since 1890

- ▶ The increase in output is not simply the result of the large increase in US population from 63 million to more than 320 million.
- ▶ Output per person has risen by a factor of 10.



Measuring the Standard of Living

- ▶ We care about growth because of the standard of living.
- ▶ Output per person (or per hour worked, or consumption per person), rather than output itself, is the variable we compare over time or across countries.
- ▶ We need to correct for variations in exchange rates and systematic differences in prices across countries.
- ▶ <https://www.youtube.com/watch?v=HEB4tvIRTXo>

Purchasing Power Parity

- ▶ Exchange rates can vary a lot
- ▶ e.g., USD increased and then decreased in the 1980s by roughly 50% against currencies of the trading partners of the United States.
- ▶ In 2018, GDP per person in India, using the current exchange rate, was \$2,016 compared to \$62,517 in US. But in India, prices of basic goods are much lower.
- ▶ In general, the lower a country's output per person, the lower the prices of food and basic services in that country.
- ▶ When comparing the standard of living across countries, we use **purchasing power parity (PPP)** numbers which adjust the numbers for the purchasing power of different countries.

The Construction of PPP Numbers: Example

- ▶ US: Each year, people buy a new car for \$10,000, and spend another \$10,000 on food.
- ▶ Russia: People spend 40,000 rubles on cars (each lasts for 15 years) a year, and 80,000 rubles on food.
- ▶ If the exchange rate is $\$1 = 60$ rubles, consumption per person in Russia is $\$2,000 = 10\%$ of US. (Now it is $\approx \$1 = 100$ rubles.)
- ▶ Using US prices for both countries, consumption per person is:
 - ▶ US: $\$10,000 + \$10,000 = \$20,000$ in the US
 - ▶ Russia: $(1/15) \times \$10,000 + \$10,000 = \$10,700$ in Russia
- ▶ Then, Russian consumption per person is 53.5% of US consumption per person.

PPP

- ▶ When comparing rich versus poor countries, the differences between numbers based on PPP and current x-rates can be large.
- ▶ At current x-rates, ratio of GDP per person in US-India = 31.3.
- ▶ Using PPP numbers, the ratio is “only” 8.
- ▶ Differences between numbers based on PPP numbers and current x-rates are typically smaller when making comparisons among rich countries.
- ▶ With x-rates in 2011, US-Germany 1.28; with PPP numbers, 1.18.
- ▶ <https://data.oecd.org/conversion/purchasing-power-parities-ppp.htm>

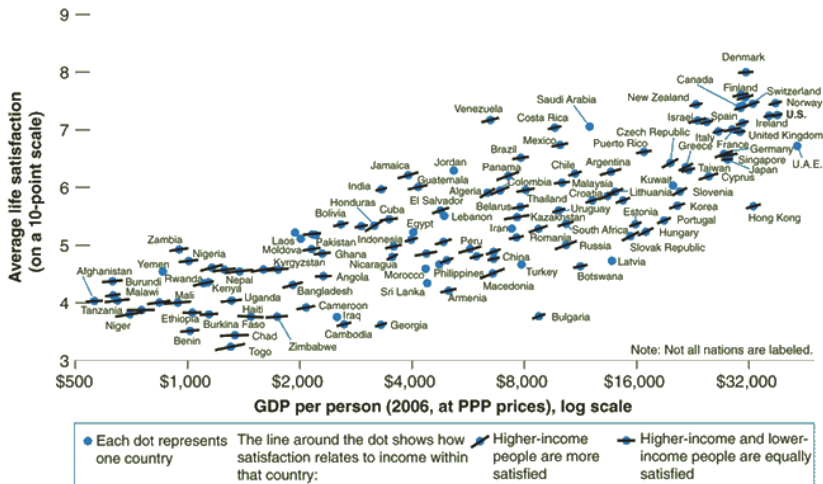
Indicators of Growth

- ▶ For households' or consumers' welfare, what matters is consumption rather than income.
- ▶ Therefore, we can use consumption per person as a measure of standard of living instead of output per person.
- ▶ However, the two are highly correlated (i.e., rankings of countries is roughly the same).
- ▶ On the production side, we may be interested in differences in productivity as opposed to the standard of living.
- ▶ Output per worker or output per hour worked may be a better measure than output per person.
 - ▶ e.g., 1,780 hours per year in US versus 1,360 in Germany (latter has higher proportion of part-time employment)

Does Money Lead to Happiness?

- ▶ Gallup World Poll survey 2006:
 - ▶ Here is a ladder representing the 'ladder of life.' Let's suppose the top of the ladder represents the best possible life for you; and the bottom, the worst possible life for you. On which step of the ladder do you feel you personally stand at the present time?
 - ▶ 0 to 10.
- ▶ <https://news.gallup.com/opinion/gallup/206468/happiest-unhappiest-countries-world.aspx>

Does Money Lead to Happiness? (con.)



Source: Betsey Stevenson and Justin Wolfers, Wharton School at the University of Pennsylvania.

Does Money Lead to Happiness? (con.)

- ▶ Strong positive correlation across countries.
- ▶ Within each country, rich people are generally happier than poor people.
- ▶ Individual happiness depends on much more than income, but it definitely increases with income.
- ▶ The idea that there is a critical level of income beyond which income no longer impacts well-being is intuitively appealing, but may be at odds with the data.
- ▶ GDP per person (level or growth) is a “decent” measure of happiness/welfare.

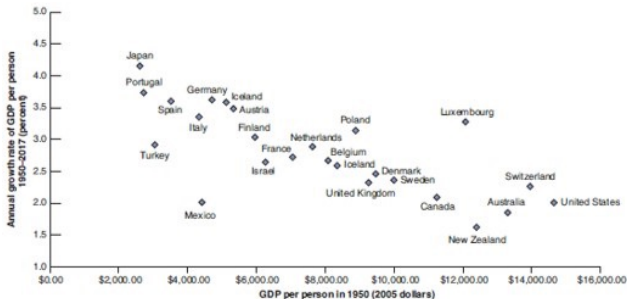
Growth in Rich Countries since 1950

- ▶ 4 countries: broadly representative of other rich countries

	Annual Growth Rate Output per Person (%)	Real Output per Person (2011 dollars)		
	1950–2017	1950	2017	2017/1950
France	2.6	7,025	39,461	5.6
Japan	4.1	2,531	40,374	15.9
United Kingdom	2.1	9,354	39,128	4.2
United States	2.0	14,569	54,995	3.8
Average	2.7	8,370	43,490	5.2

- ▶ Large increase in output per person due in part to compounding.
 - ▶ e.g., permanent increase in 1%: after 40 years, 48% higher income
- ▶ Convergence of output per person across countries
 - ▶ Countries that were behind have grown faster, reducing the gap with the US

Convergence of OECD Countries



- ▶ Countries with lower levels of output per person in 1950 have typically grown faster. But these are the ones that made it!
- ▶ How general is convergence?
 - ▶ e.g., Uruguay and Venezuela had the same output per person as France in 1950, but today 1/3 and 1/5.

UK vs Ireland



- ▶ GDP per capita, 2017 international dollars
- ▶ By 2019, Ireland 1.86 times richer

A Broader Look across Time and Space

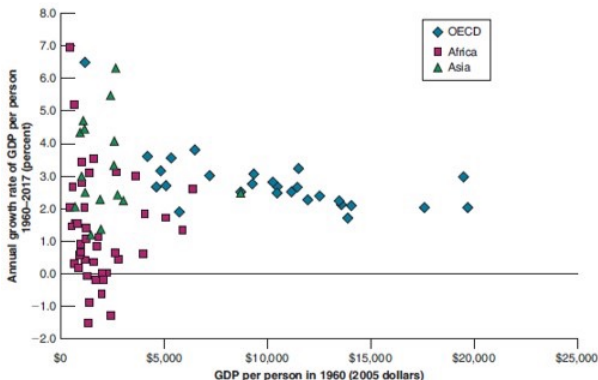
- ▶ From the end of the Roman Empire in 476 to roughly year 1500, Europe was in a “**Malthusian trap**”
 - ▶ Stagnation of output per person: most workers were in agriculture with little technological progress.
 - ▶ Although there was some output growth, a roughly proportional increase in population led to roughly constant output per person.
 - ▶ Malthus argued that any increase in output would lead to a decrease in mortality, leading to an increase in population until output per person was back to its initial level.
- ▶ After 1500, growth of output per person turned positive but still small (0.1% per year)
- ▶ Between 1820 and 1950 (Industrial Revolution), US growth was still 1.5% per year. $\Rightarrow$ Sustained growth is a recent phenomenon!

Industrial Revolutions

- ▶ Industrial Revolution, 1760-1840: machines, new chemical and iron production, steam power, water power (e.g., production of textiles in factories)
- ▶ Industry 2.0, 1870-1914: telegraph, oil, electricity and electrical machines, modern assembly/production line (i.e., mass production)
- ▶ Industry 3.0, 1970-present: computers, mobile, software, internet, digitalization and automation
- ▶ Industry 4.0, now: robots, big data, cloud computing, artificial intelligence, 3D printing, smart systems

Growth Across Countries

- A look at 85 countries:



- There is no clear relation.

More Details

- ▶ For the OECD countries, there is clear evidence of convergence.
- ▶ Convergence is also visible for many Asian countries, especially for those with high growth rates, called the four tigers: Singapore, Taiwan, Hong Kong, and South Korea.
- ▶ More recently, the major story has been China, both because of its very high growth rates and because of its sheer size.
 - ▶ Over the period 1960-2017, growth of output per person in China has been 4.5% per year on average.
 - ▶ But, because it started low, its output per person is still only about one-fourth of US.
- ▶ Most African countries were very poor in 1960, and some even had negative growth of output per person between 1960-2017 (e.g., internal, external conflicts). Hope? 1.3% in 1990s, 3% since 2000

Production Functions

- ▶ Aggregate production function F (previously, $Y = N$):

$$Y = F(K, N)$$

- ▶ where Y is output, K is capital, and N is labor.
- ▶ **Constant returns to scale:**

$$F(xK, xN) = xY$$

- ▶ **Decreasing returns to capital:** Increases in K lead to smaller and smaller increases in Y (i.e., diminishing marginal returns, second derivative < 0)
- ▶ **Decreasing returns to labor:** Increases in N lead to smaller and smaller increases in Y .

Sources of Growth

- ▶ The production function and constant returns to scale imply a simple relation between Y/N and K/N :

$$\frac{Y}{N} = F\left(\frac{K}{N}, \frac{N}{N}\right) = F\left(\frac{K}{N}, 1\right)$$

- ▶ The function F depends on the state of technology: more advanced technology $\Rightarrow$ more Y from same quantity of K and N .
 - ▶ Increases in K/N : Movements along the production function.
 - ▶ Improvements in the state of technology: Shifts (up) of the production function.
- ▶ Potential sources of growth:
 - ▶ Capital accumulation (a higher saving rate)
 - ▶ Technological progress (the improvement in the state of technology).

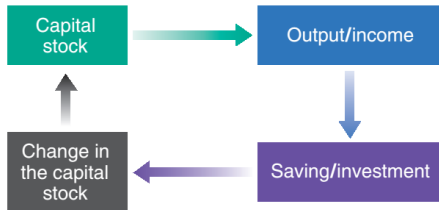
Graphical Representation

Saving, Capital Accumulation, and Output

- ▶ Since 1970, the US saving ratio (i.e., ratio of saving to gross domestic product) has averaged only 17%.
- ▶ Compared to 23% in Germany and 29% in Japan.
- ▶ China (2012):
<https://www.youtube.com/watch?v=7qKhSSkUbt8>
- ▶ Would increasing the saving rate lead to sustained higher growth in the future?
- ▶ Growth rate versus level: an increase in the saving rate would lead to higher growth for some time and eventually to a higher standard of living.
- ▶ But not sustained growth.

Interactions between Output and Capital

- Output in the long run depends on two relations:
 1. The amount of capital determines the amount of output.
 2. Amount of output being produced determines amount of saving. This in turn determines amount of capital accumulated over time.



Effects of Capital on Output

- ▶ Recall:

$$\frac{Y}{N} = F\left(\frac{K}{N}, 1\right) \quad \text{or} \quad f\left(\frac{K}{N}\right) \equiv F\left(\frac{K}{N}, 1\right)$$

- ▶ Assume that N is constant, and there is no technological progress, so f does not change over time:

$$\frac{Y_t}{N} = f\left(\frac{K_t}{N}\right)$$

- ▶ We focus on the role of capital accumulation.
- ▶ Higher capital per worker leads to higher output per worker.

Output and Investment

► Assume:

- The economy is closed: $I = S + (T - G)$
- Public saving $T - G$ is 0. $I = S$
- Private saving is proportional to income: $S = sY$

► So the relation between output and investment:

$$I_t = sY_t$$

- Investment is proportional to output.
- The higher (lower) output is, the higher (lower) is saving and so the higher (lower) is investment.

Investment and Capital Accumulation

- ▶ The evolution of the capital stock is:

$$K_{t+1} = (1 - \delta)K_t + I_t$$

- ▶ Replace investment and divide both sides by N :

$$\frac{K_{t+1}}{N} = (1 - \delta)\frac{K_t}{N} + s\frac{Y_t}{N}$$

- ▶ or

$$\frac{K_{t+1}}{N} - \frac{K_t}{N} = s\frac{Y_t}{N} - \delta\frac{K_t}{N}$$

- ▶ The change in the capital stock per worker is equal to saving per worker minus depreciation.

Dynamics of Capital and Output

- ▶ Two equations:

$$\text{Production side: } \frac{Y_t}{N} = f\left(\frac{K_t}{N}\right)$$

$$\text{Saving side: } \frac{K_{t+1}}{N} - \frac{K_t}{N} = s \frac{Y_t}{N} - \delta \frac{K_t}{N}$$

- ▶ Combining them:

$$\frac{K_{t+1}}{N} - \frac{K_t}{N} = sf\left(\frac{K_t}{N}\right) - \delta \frac{K_t}{N}$$

Change in capital Investment – depreciation
year t to $t + 1$ during year t during year t

- ▶ If investment per worker exceeds (is less than) depreciation per worker, the change in capital per worker is positive (negative).

Dynamics of Capital and Output (con.)

Steady State

- ▶ The state in which output per worker and capital per worker are no longer changing is called the **steady state** of the economy.

$$sf\left(\frac{K^*}{N}\right) = \delta \frac{K^*}{N}$$

- ▶ The steady-state value of capital per worker is such that the amount of saving per worker is sufficient to cover depreciation of the capital stock per worker.
- ▶ Steady-state output per worker:

$$\frac{Y^*}{N} = f\left(\frac{K^*}{N}\right)$$

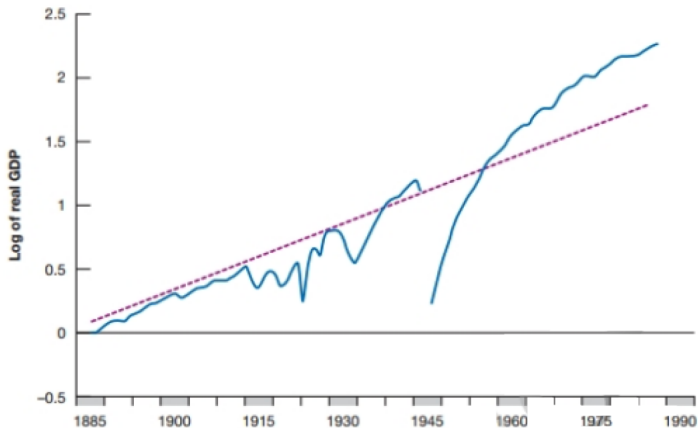
Capital Accumulation and Growth in France

- ▶ France suffered heavy losses in capital when WWII ended (1945).
- ▶ The growth model predicts that France would experience high capital accumulation and output growth for some time.
- ▶ Proportion of French capital stock destroyed by end of WWII:

Railways	Tracks	6%	Rivers	Waterways	86%
	Stations	38%		Canal locks	11%
	Engines	21%		Barges	80%
	Hardware	60%	Buildings	(numbers)	
Roads	Cars	31%		Dwellings	1,229,000
	Trucks	40%		Industrial	246,000

- ▶ From 1946-50, real GDP grew at 9.6% per year: 60% over 5 years!
- ▶ Also, more modern capital and production techniques introduced.

The German Experience



Saving Rate and Output

- ▶ The saving rate has no effect on the *long-run* growth rate of output per worker, which is equal to zero.
 - ▶ Eventually, the economy converges to a constant level of output per worker.
- ▶ The saving rate determines the level of output per worker in the long run.
 - ▶ All else equal, a higher saving rate increases output per worker in the long run.
- ▶ An increase in the saving rate will lead to higher growth of output per worker for some time, but not forever.
 - ▶ This follows from the two propositions above. The economy must go through a period of positive growth.

The Effects of Different Saving Rates

Saving Rate and Consumption

- ▶ What matters to people is not how many is produced, but how much they consume.
- ▶ Governments can affect the saving rate by:
 - ▶ Changing public saving (i.e., running a budget surplus)
 - ▶ Using taxes to affect private saving (e.g., tax break on savings)
- ▶ If saving rate is zero, capital + output + consumption are all zero.
- ▶ If saving rate is one, all income is saved and consumption is zero (all output devoted to replacing depreciation).
- ▶ There must be some intermediate saving rate that maximizes the steady-state level of consumption: **Golden-rule level of capital.**

Golden-Rule Level of Capital

- ▶ The level of capital associated with the saving rate s_G that yields the highest level of consumption in steady state. Increases in saving
 - ▶ below this value decrease consumption initially, but increases consumption in the long run.
 - ▶ above this value decrease consumption both initially and in the long run. (Increase in output from capital accumulation too small to cover depreciation.)

Golden-Rule Level of Capital (con.)

- ▶ For $s \in [0, s_G]$, a higher saving rate leads to higher capital per worker, higher output per worker and consumption per worker.
- ▶ For $s > s_G$, a higher saving rate still leads to higher capital per worker and output per worker, but lower consumption per worker.
- ▶ Empirically, most OECD countries are actually far below their golden-rule level of capital.
- ▶ Trade-off: An increase in the saving rate leads to lower consumption for some time but higher consumption later.
- ▶ Should governments care about welfare of current or future generations? Who votes?

Ageing, Retirement, and Slow Growth in Japan

- ▶ A combination of financial troubles and insufficient effective demand led to a very sluggish growth for Japan in the early 1990s.
- ▶ Zombie banks: insolvent but operating because of government policies/support
- ▶ <https://www.youtube.com/watch?v=TXgKI6Kk0gQ>
- ▶ Financial troubles (e.g., problem loans at banks) overcome in early 2000s, but GDP continued to grow slowly

Ageing, Retirement, and Slow Growth in Japan (con.)

- ▶ Japan is among the fastest ageing populations in the world.
- ▶ It also sets a low mandatory retirement age for many of its workers, often years before the age required to benefit from state or national pensions.
- ▶ Early retirees with no access to state or national pensions must draw on their savings.
- ▶ In 1994 (“lost decade”), Japanese households saved around 11.56% of their disposable income (compared to US with 6.53%).
- ▶ But it reached an all-time low of -0.47% in 2014.

The Cobb-Douglas Production Function

- ▶ Economists often use the Cobb-Douglas production function:

$$Y = K^{\alpha} N^{1-\alpha}$$

- ▶ This function gives a good description of the relation between output, physical capital, and labor in the US from 1899 to 1922.
- ▶ In steady state, saving per worker must be equal to depreciation per worker, implying that:

$$s(K^*/N)^{\alpha} = \delta(K^*/N)$$

- ▶ where K^* is the steady-state level of capital.

The Cobb-Douglas Production Function (con.)

- Rewrite as:

$$s = \delta(K^*/N)^{1-\alpha}$$

- The steady-state level of capital and output per worker are:

$$(K^*/N) = (s/\delta)^{1/(1-\alpha)}$$

$$(Y^*/N) = (K^*/N)^\alpha = (s/\delta)^{\alpha/(1-\alpha)}$$

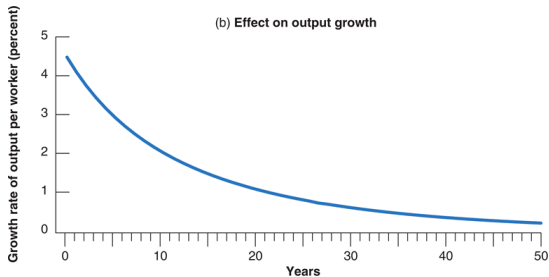
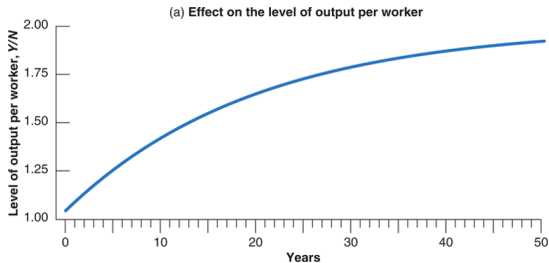
- If $\alpha = 0.5$, then:

$$(K^*/N) = (s/\delta)^2, \text{ and } (Y^*/N) = (s/\delta)$$

- which implies that a doubling of the saving rate leads to a doubling in steady-state output per worker.

Dynamic Effects of an Increase in the Saving Rate

Dynamic Effects (con.)



The US Growth Rate

- ▶ Can the low saving/investment rate explain why the US growth rate has been so low, relative to other OECD countries, since 1950?
- ▶ Yes *if*
 - ▶ US had higher saving rate in the past
 - ▶ And if the saving rate had fallen substantially in the last 70 years
- ▶ But US saving rate has been low for long time, therefore, this is not a good explanation.

Saving Rate and the Golden Rule

Saving Rate and the Golden Rule

- ▶ Assuming a depreciation rate of 10%

Saving Rate s	Capital per Worker K/N	Output per Worker Y/N	Consumption per Worker C/N
0.0	0.0	0.0	0.0
0.1	1.0	1.0	0.9
0.2	4.0	2.0	1.6
0.3	9.0	3.0	2.1
0.4	16.0	4.0	2.4
0.5	25.0	5.0	2.5
0.6	36.0	6.0	2.4
...	...	...	...
1.0	100.0	10.0	0.0

Physical versus Human Capital

- ▶ Human capital (H): The set of skills of the workers in the economy built through education and on-the-job training.
- ▶ The increase in human capital has been as large as the increase in physical capital over the last two centuries.
- ▶ At the beginning of the Industrial Revolution (1760-1840), only 30% of today's OECD countries could read. Today, it is 95%.
- ▶ In OECD countries, 100% of children get a primary education, 90% secondary, 38% higher.
- ▶ Poor countries (GDP per capita below \$400), 95%, 32%, 4%.

Extending the Production Function

- ▶ The production function with human capital (diminishing returns):

$$\frac{Y}{N} = f\left(\frac{K}{N}, \frac{H}{N}\right)$$

- ▶ Relative importance? US spends 6.5% of GDP on formal education, gross investment rate for physical capital is 16%.
 - ▶ Higher education is both consumption and investment.
 - ▶ Actual cost of post-secondary education does not include opportunity cost.
 - ▶ On-the job training (formal or informal) is not included.
 - ▶ Skills deteriorate too, but more slowly.
- ▶ Recent studies conclude that investment in physical capital and in education play roughly similar roles in the determination of output.

Endogenous Growth

- ▶ Production functions feature decreasing returns to human/physical capital, so sustained growth only with steady increase in capital and skilled workers.
- ▶ The current consensus is that *given the rate of technological progress*, higher rates of saving or spending on education do not lead to a permanently higher growth rate, only the level.
- ▶ Capital accumulation cannot by itself sustain growth, it requires technological progress.
- ▶ Might technological progress depend on the level of human capital in the economy?
- ▶ Models of **endogenous growth**: Steady-state growth in output per worker depends on variables such as the saving rate and the rate of spending on education, even without technological progress.

Paul Romer

- ▶ Nobel Prize in Economics 2018
- ▶ Prize motivation: “for integrating technological innovations into long-run macroeconomic analysis.”
- ▶ At its heart, economics deals with the management of scarce resources. Nature dictates the main constraints on economic growth and our knowledge determines how well we deal with these constraints. Paul Romer has demonstrated how knowledge can function as a driver of long-term economic growth. He showed how economic forces govern the willingness of firms to produce new ideas. Romer’s central theory, which was published in 1990, explains how ideas are different to other goods and require specific conditions to thrive in a market.